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Etch of nano-TSV with smooth sidewall and excellent selection ratio for backside power delivery network

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ABSTRACT

Backside Power Delivery Network (BSPDN) is a crucial technology for integrated circuits at sub-3 nm technology nodes. The primary challenge resides in utilizing nano through silicon via (nano-TSV) to establish connections between the backside power network and buried power rails, thereby facilitating transistor powering. The key technology is to ensure a smooth sidewall morphology and prevent damage to buried power rails (BPR) due to over-etching. In this study, non-Bosch and Bosch techniques are compared using simulation. The results demonstrate that while the non-Bosch technique yields smooth sidewalls, it inevitably leads to over-etching, whereas Bosch effectively avoids over-etching. The etching of scallop-free nano-TSV is achieved by optimizing the Bosch process, which involves the use of inductively coupled plasma (ICP). Finally, metal filling of nano-TSV is successfully achieved. Thus, the nano-TSV etching method is established as viable for BSPDN.

1. Introduction

In large-scale integration circuits, the pursuit of 2D scaling to enhance power and performance while simultaneously reducing area faces significant challenges, including routing obstruction and back-end of line (BEOL) congestion [1]. The Backside Power Delivery Network (BSPDN) involves separating the PDN from the signal routing and implementing the PDN on the backside of the device wafer. It supplies power to the transistors through nano-TSVs connected to buried power rails (BPR) [2–6]. *E*-Core implementation in Intel 4 with BSPDN reduces signal routing constraints, resulting in a logic cell density of 90 % for large-area designs. This implementation increases the maximum CPU frequency by 5 %, and reduces the IR drop to 1/5 [7,8]. For ARM™ at the A14 node using BSPDN, the frequency improves by 6 % compared to the frontside PDN, stemming from a 16 % reduction in core area and a 23 % reduction in dynamic IR drop [4].

Nevertheless, the scallop pattern resulting from the Bosch etch process is detrimental to nanoscale TSV [9–13]. Thermal stress at the TaN/SiO_2 interface for nanoscale TSVs with the scallop is as high as 581 MPa when heating to 120 °C. Conversely, when nano-TSV exhibits smooth sidewall, the stress is approximately half of that. High stress may lead to the crack propagation of dielectric layer and potential large current leakage. Furthermore, large crack may happen between two TSV and the using life of electrical product will be lower [9,14]. Furthermore, the scallop pattern also complicates subsequent metal filling more challenging. The nano-TSV for BSPDN will be connected to the BPR as shown in Fig. 1. It is imperative that the etching process should not result in damage to the BPR, otherwise this will lead to resistance increase. Thus, a smooth-sidewall etching process must be investigated for nano-TSV, as well as mitigating BRP damage.

In this study, the effect of nano-TSV scallop patterns on the metal filling is investigated. Non-Bosch and Bosch techniques are compared using simulation and inductively coupled plasma (ICP). Both etching methods achieve the smooth sidewall of nano-TSV. The optimized Bosch

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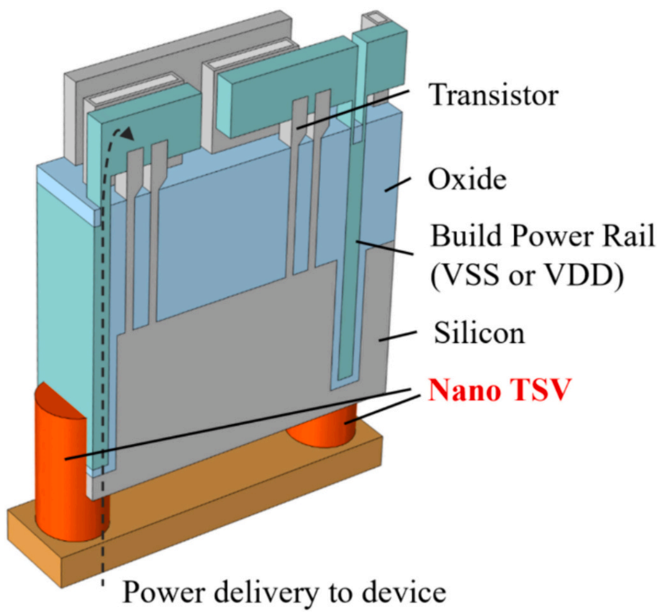

Fig. 1. Illustration of the BSPDN with nano-TSV.

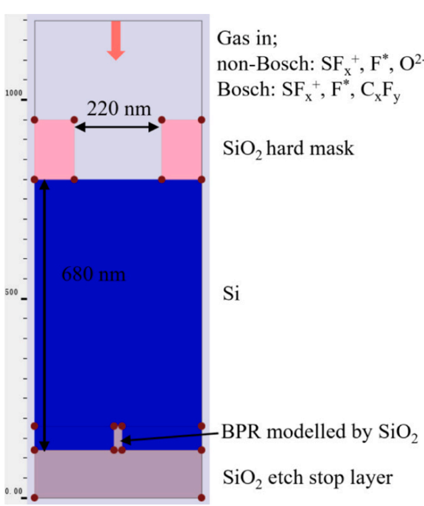

Fig. 2. Etching simulation models for non-Bosch and Bosch.

process effectively prevents over-etch, thereby averting damage to the BPR within the SiO_2 stop layer. Finally, the metal filling of the nano-TSV is completed by the electroplating process.

2. Experiment

The 2 cm × 2 cm SOI chip is employed as the device chip, with embedded silicon oxide serving as the isolation layer under the BPR. Following SiO_2 deposition by PECVD as the hard mask, PMMA 950 K photoresist is spin-coated, and the nano via arrays are patterned using electron beam lithography (EBL). Subsequently, the SiO_2 hard mask is etched by reaction ion etching (RIE). Both non-Bosch and Bosch etching techniques are then performed using inductively coupled plasma etching (ICP). In the non-Bosch process, the etching under varying power and bias voltages is studied. In the Bosch etching process, the effects of bias voltage and gas flow rate on the etch rate and scallop width are examined separately. The bias voltages are set at 15 W, 25 W, 35 W, and 45 W, while the SF_6 flow rates are set at 90 sccm, 70 sccm, 30 sccm, and 20 sccm. After depositing the insulating and barrier layers using plasma enhanced atomic layer deposition (PEALD), the seed layer is deposited using PVD. The final metal filling is then carried out using electroplating. The nano-TSV profiles are produced using Focused Ion Beam (FIB), and Scanning Electron Microscope (SEM) is used to observe the nano-TSV morphology and over-etching.

In order to investigate the effect of non-Bosch and Bosch on over-etching, a model with a diameter of 220 nm and an etching depth of 680 nm was constructed using PEGASUS software as shown in Fig. 2. The etching process proceeds until it reaches the bottom etch stop layer. The outermost layer of the BPR is the SiO_2 etch stop layer, which is modeled in this simulation by protruding SiO_2 layer. During the etching process, ions and F* flow in from the top surface, with the energy and angle of incident particles following a Maxwell distribution. In the non-Bosch etching process, SF_x^+, F*, and O^{2-} ions enter simultaneously. The primary reaction equations are as follows:

$$Si_{(s)} + O_{(g)} -> SiO_{2(s)} \tag{1}$$

$$Si_{(s)} + F^*_{(g)} -> SiF_{4(s)} \tag{2}$$

$$SiO_{2(s)} + Ion_{(g)} -> SiO_{2(g)} \tag{3}$$

$O_{(g)}$ reacts with Si to form $SiO_{2(s)}$. This $SiO_{2(s)}$ is then sputtered by ions with energy and direction, resulting in the formation of $SiO_{2(g)}$. During the Bosch etching process, SF_x^+, F*, and C_xF_y ions alternate in entering. The primary reaction equations during the deposition step are as follows:

$$Si(s) + C_xF_y(g) -> Polymer(s) \tag{4}$$

The primary reaction equations during the etch step are as follows:

$$Si_{(s)} + 4F_{(g)} = SiF_{4(g)} \tag{5}$$

$$Polymer_{(s)} + Ion = CF_{z(g)} + SF_{x(g)} \tag{6}$$

$$SiO_{2(s)} + Ion_{(g)} -> SiO_{2(g)} \tag{7}$$

3. Results and discussion

In our previous study, the effect of scallop patterns on thermal stress has been analyzed [9]. The present research examines the impact of scallop morphology on the metal filling. In the TSV filling process, the widely used method is electroplating, which is applied after the seed layer deposition by PVD. Our subsequent study presents that the scallop has a significant influence on the filling quality in nano-TSV. Fig. 3 (a) shows the nano-TSV etched by Bosch process with the diameter of 220 nm. The width of scallop pattern is 33 nm, which is about 1/7 of the diameter. Subsequently, A seed layer is then deposited by PVD. Fig. 3 (b) illustrates the seed layer exhibits a clear crack in the middle location of nano-TSV, which is unfavorable for conductivity. This causes significant voids after the metal filling process, as illustrated in Fig. 3 (c) and (d). Therefore, the etching of nano-TSV with smooth sidewall is very important. Then non-Bosch and Bosch etching technology are designed and evaluated to obtain scallop sidewall as below.

(a) Nano-TSV filling

Non-Bosch etching can be employed to produce nano-TSV with smooth sidewall. The etching process involves F*, Cl*, etc. in conjunction with oxygen (O), resulting in anisotropic etching. The etching

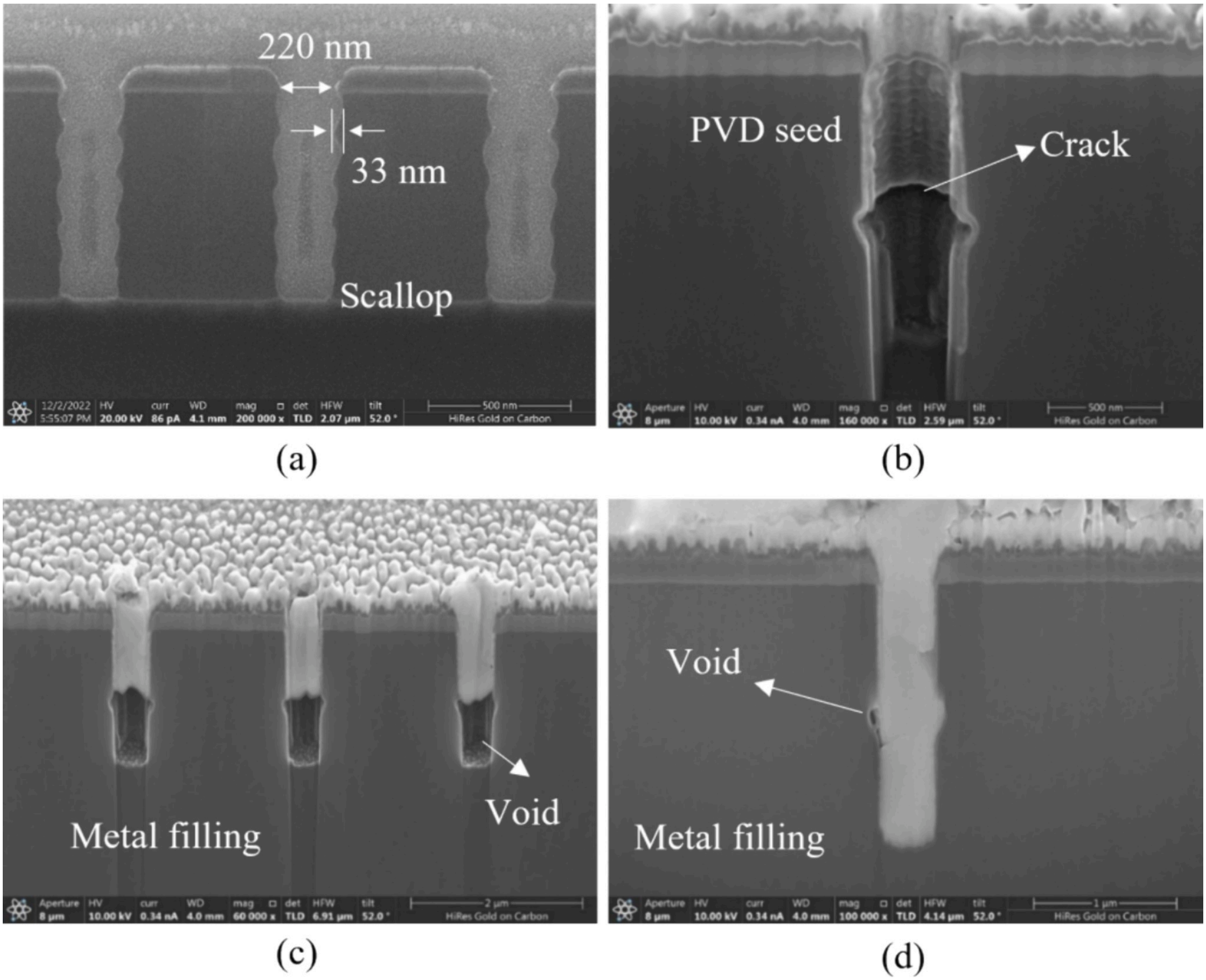

Fig. 3. Scallop morphology in different nano-TSV fabrication process: (a) Si etching, (b) seed layer deposition by PVD, (c) and (d) voids after Cu electroplating.

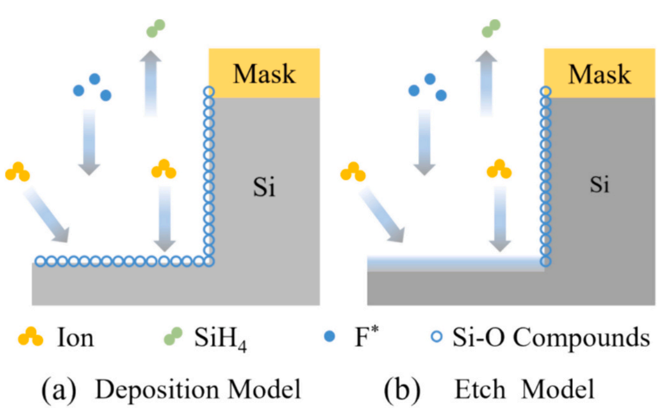

Fig. 4. The anisotropic etching mechanism for non-Bosch process.

Table I
The common bond dissociation energy.

Bond Dissociation Energy	Si-O	Si-F	Si-Cl	Si-Br	Si-Si	Si-I
(kcal/mol)	191 ± 3.2	102.4 ± 4	99.6 ± 1.5	85.6 ± 2	74.1	58.1 ± 2

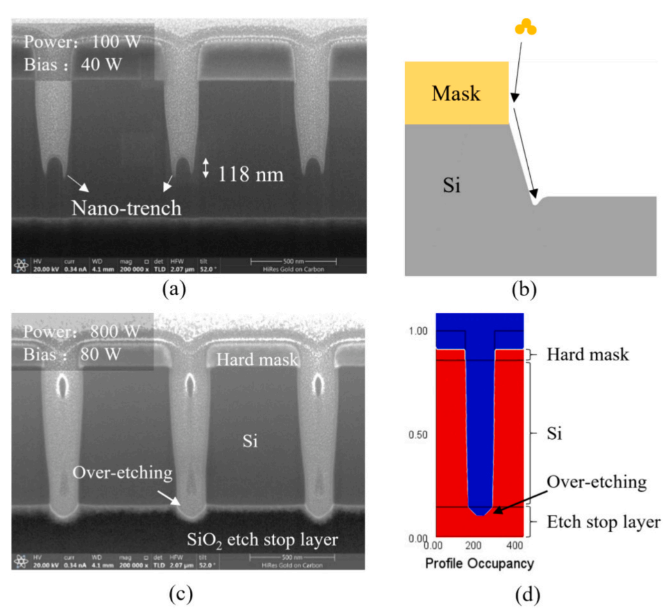

Fig. 5. Micro-trench effect on silicon wafer (a) and formation mechanism (b) in the nano-TSV etching. Etching results for SOI wafer (c) and simulation results (d) after increasing power and bias.

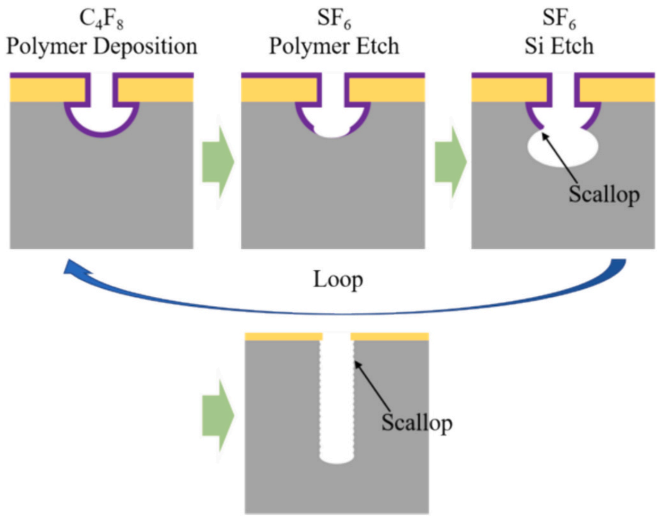

Fig. 6. Schematic diagram of the Bosch Process for Silicon etching.

mechanism is illustrated in Fig. 4. In the non-Bosch etching process, deposition mode and etch mode are carried out simultaneously. The common bond dissociation energy are shown in Table I, and Si—O is 191 ± 3.2 kcal/mol, which is much larger than Si-X (X is F, Cl, Br, I). In the deposition mode, larger Si—O dissociation energy makes it more stable and proves to deposit Si—O compounds on the sidewall and bottom of nano-TSV as shown in Fig. 4 (a). This protects the Si from corrosion by F* and Cl*. In the etching mode, ions under the vertical electric field are directed onto the bottom of nano-TSV, thereby breaking the Si—O at the bottom. This allows Si to react with F* or Cl* to form volatile SiX_4, which is then pumped away to achieve anisotropic etching.

(b) Non-Bosch for nano-TSV etching

The morphology of non-Bosch etching is illustrated in Fig. 5(a). The bottom of the TSV demonstrates the phenomenon of fast etching speed at the edge and slow etching speed in the middle, which is referred to as a nano-trench. This etching profile has profoundly deleterious consequences. It is due to the fact that a large number of non-vertical ions are swept into the bottom edge of the sidewall. Thus, the ions aggregated at the bottom edge lead to an additional increase in the etching rate at the sidewall corners as shown in Fig. 5 (b), which is consistent with reference [15]. It has been reported that increasing the energy of incident ions can suppress the nano-trench effect [16]. Because the ion directionality will be more vertical when the incident ion energy is high. As a result, there is a decrease in the swept-incidence accumulation of ions. It is possible to significantly increase the ion energy by increasing the power and bias voltage. Therefore, higher power and bias voltage are designed and studied [16]. Figs. 5 (a) and (c) illustrates the etching outcome with the power increasing from 100 W to 800 W and the bias from 40 W to 80 W. The nano-trench phenomenon is no longer observed. However, due to the elevated ion energy, the over-etching phenomenon emerges. It becomes a challenge for the etching process to halt at the etch-stop layer. This results in damage to the BPR in the BSPDN process, leading to the resistance increase. Then, the process of non-Bosch etching is simulated by PEGASUS software to verify the assumption above. The simulation results also demonstrate a significant over-etching phenomenon happens when higher power and bias are adopted.

(C) Bosch for nano-TSV etching

The Bosch etching process differs from non-Bosch technology in that the etching and passivation processes occur separately, as illustrated in Fig. 6. The Bosch process is a continuously looping sequence of three steps: polymer deposition, polymer etching and silicon etching. During the process, loops are established between the etching step using SF_6 and the polymer deposition step using C_4F_8, resulting in the formation of scallop patterns on the sidewalls and the creation of a rough surface [17,18]. The Bosch technology has a much lower ion energy in the etching step compared to non-Bosch, avoiding over-etching and the destruction of the BPR.

Table II
Base etching process conditions.

Process variable	Base value	Range studied
Power	600 W	
Bias	35 W	15–45 W
Temperature	25 °C	
C_4F_8 flow	90 sccm	
SF_6 flow	90 sccm	20–90 sccm
Pressure	Etch: 30 mTorr Depositon: 20 mTorr	

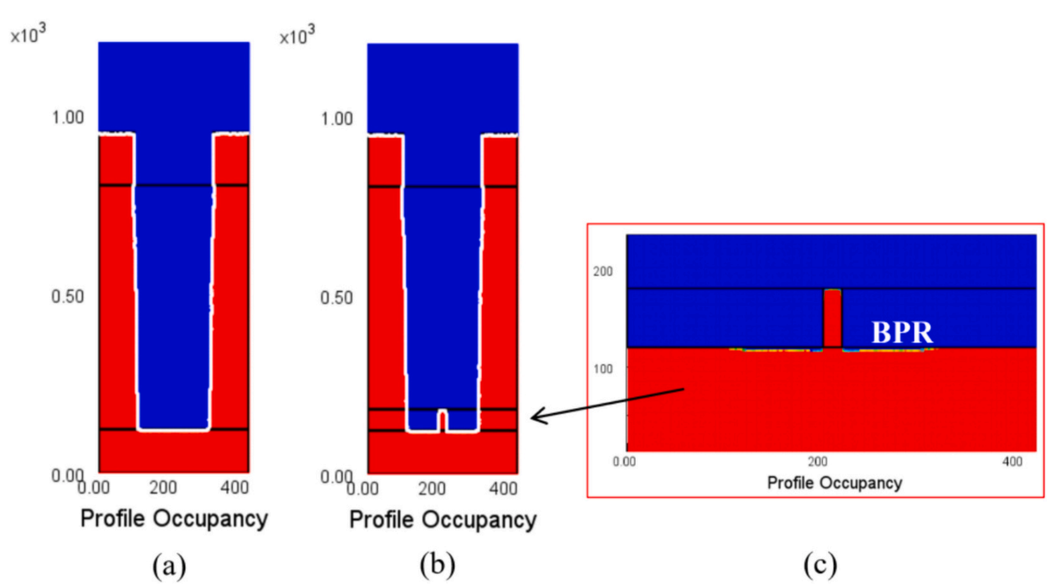

Fig. 7. (a) Software simulation of the Bosch process, (b) Simulation of etching results with BPR and (c) partial enlargement.

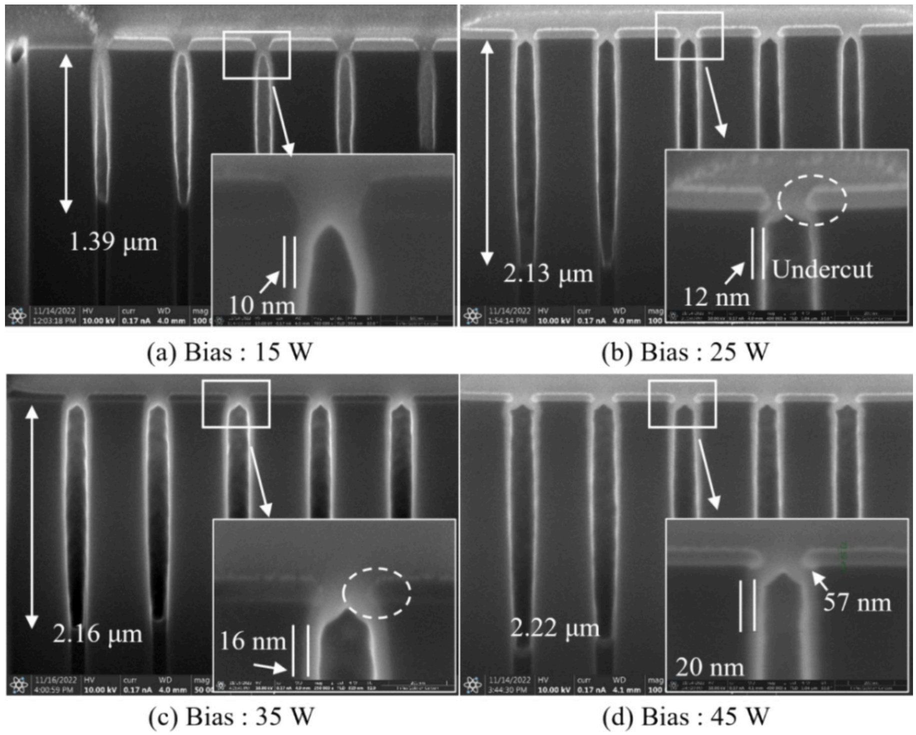

Fig. 8. SEM cross-sections of nano-TSV etched under different bias voltage.

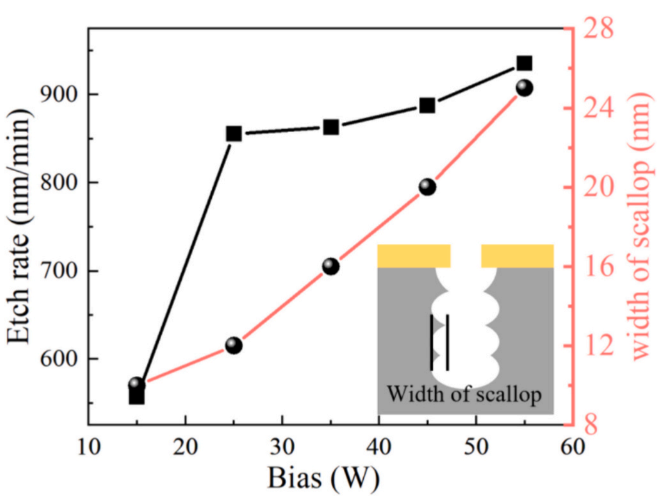

Fig. 9. Bias voltage versus etch rate and scallop width.

In order to prevent the formation of conspicuous scallops, the etching rate is slowed down by reducing the SF_6 flow rate, power, and bias parameter. The etching process is dependent upon the reaction of F* with Si to form SiF_4. A reduction in the flow rate and power of SF_6 can result in a reduction in the concentration of F* and a slowing down of the etching rate. A reduction in the bias voltage can result in a reduction in the ion energy, a reduction in the sputtering effect of ions in the etching process, and an improvement in the etching selectivity ratio. The Bosch process is simulated using PEGASUS software. Low F* flux and ion energy made the nano-TSV sidewalls smooth and without over-etching, as shown in Fig. 7(a). Fig. 7(b) shows the simulation results of etching nano-TSV with BPR. The BPR remains intact, preventing the observed resistance increase. The optimized Bosch process is suitable for use in BSPDN technology.

We systematically investigated the effects of bias and SF_6 flow on the etching rate and width of scallop in nano-TSV by adjusting these variables about their base values listed in Table II. Fig. 8 presents the etching results of nano-TSV under different bias voltage conditions, all achieving nano-TSV etching. Etching is carried out at an ICP coil power of 600 W and SF_6 flow of 90 sccm. The bias increased from 15 W to 45 W. When the bias voltage increases to 25 W, lateral etching appears below the hard mask, known as undercut. As the bias voltage continues to increase, the undercut phenomenon intensifies. At a bias voltage of 45 W, the width of the undercut reaches 57 nm. Fig. 9 illustrates the relationship curves between bias voltage, etching rate, and scallop width. With increasing bias voltage, the etching rate significantly accelerates, and the scallop width gradually increases. Bias voltage significantly influences ion energy. During the etching step, higher bias voltage results in higher ion energy, making sputtering effects more pronounced, thereby leading to a notable increase in etching rate. High-energy ions are more likely to reflect onto the sidewalls, increasing the occurrence of lateral etching. To achieve scallop-free nano-TSV, etching should be performed under low bias voltage conditions.

Fig. 10 shows the results of etching nano-TSV under different SF_6 flow. Etching is carried out at a bias of 35 W and C_4F_8 flow of 90 sccm. The SF_6 flow increased from 20 sccm to 90 sccm. Nano-TSV can be achieved under different flow conditions. Fig. 11 shows a plot of SF_6 flow versus etch rate and scallop width. As the SF_6 flow rate decreases, the etching rate markedly decreases, and the scallop width gradually reduces. At an SF_6 flow rate of 90 sccm, the etching rate is 900 nm/min, and the scallop width is 52 nm. When the SF_6 flow decreases to 20 sccm, the etching rate is 332 nm/min, with virtually no visible scallop pattern. The flow of SF_6 significantly influences the concentration of F*. Higher SF_6 flow result in increased F* concentrations, which in turn significantly enhance the etching rate. Ultimately, we successfully etch nano-TSV with smooth sidewalls and without over-etching on SOI, as shown in Fig. 12.

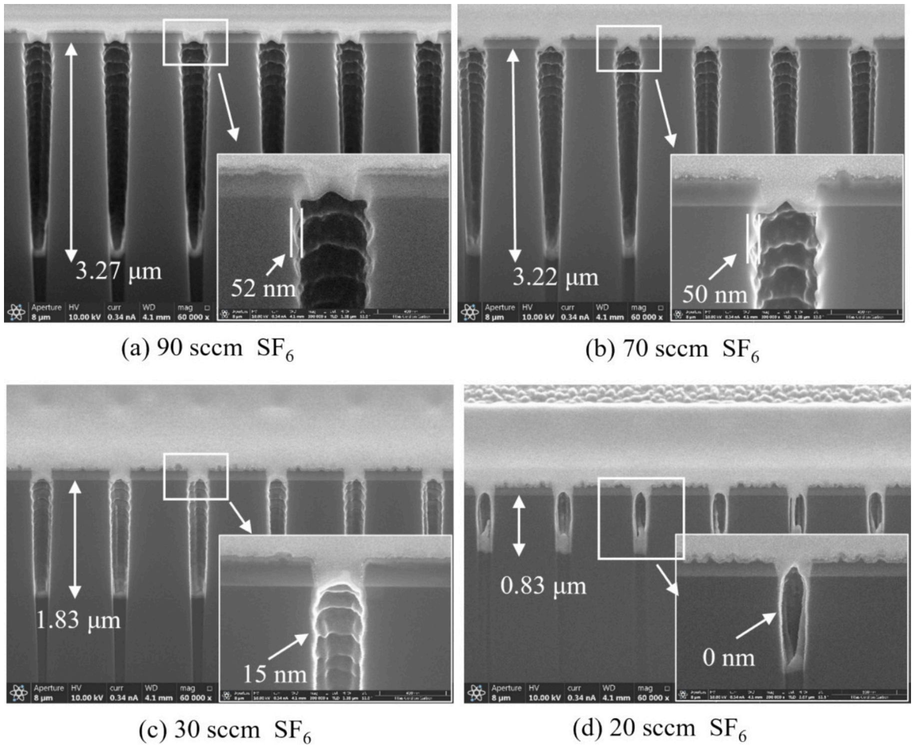

Fig. 10. SEM cross-sections of nano-TSV etched under different SF_6 flow.

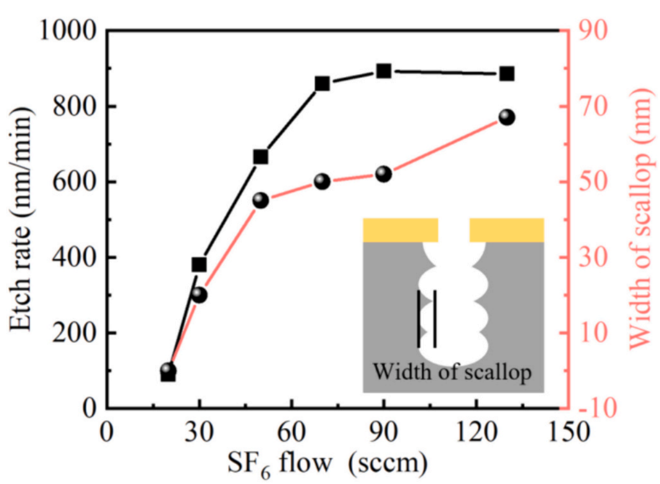

Fig. 11. SF_6 flow versus etch rate and scallop width.

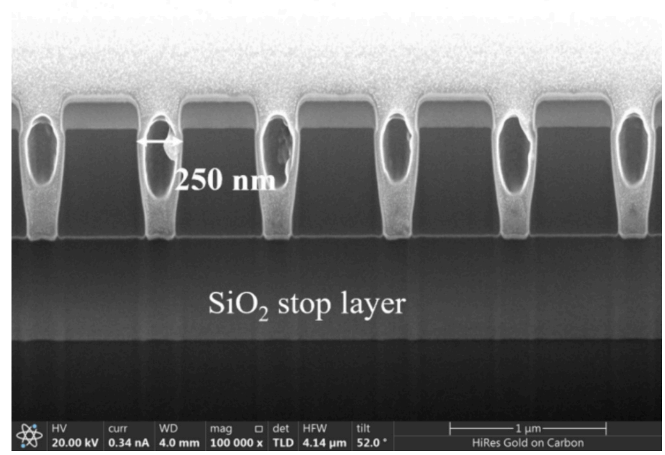

Fig. 12. (a) The result of nano-TSV etching process on SOI wafer.

Different diameters of nano-TSVs are etched using the optimized Bosch technique as shown in Fig. 13. The resulting structures exhibit excellent perpendicularity and smooth etched sidewalls, indicating that the process parameters are suitable for the etching of nano-TSVs of different diameters. This technology has the potential to be applied not only in backside-powered network technology but also in chip 3D IC, 3D NAND, CIS and other related fields. After depositing a 10 nm thick ALD TaN barrier layer and a 10 nm thick PVD Cu seed layer, Cu electroplating is performed using Shanghai Sinyang SYS^{T}2520 series copper methylsul fonate electrolyte. The electroplating process keeps the plating solution flowing and increases the diffusion rate of various ions in the solution. Cu plating is performed at 0.1 ASD. The results of the metal filling process for nano-TSV using the optimized Bosch technique are presented in Fig. 14. The void-free filling of nano-TSV with a diameter of 250 nm and a depth of 1.46 μm is successfully achieved.

4. Conclusion

In conclusion, we analyze the difficulties of the subsequent metal filling process caused by the scallop pattern. The mechanisms of both non-Bosch and Bosch etching methods are investigated using simulation and ICP. It is determined that the utilization of non-Bosch facilitates the nano-TSV etching process with smooth sidewalls, albeit resulting in inevitable damage to the BPR in the BSPDN. The ability to avoid over-etching is due to the lower ionic energy of Bosch etching compared to non-Bosch. Bias voltage and SF_6 flow significantly influence etching morphology and rate. Reducing the bias voltage slows the etching rate and decreases the width of the undercut. Decreasing the SF_6 gas flow significantly reduces the width of the scallop pattern, thereby achieving scallop-free etching. Ultimately, the scallop-free nano-TSV is achieved without causing BPR damage, using Bosch process. Subsequently, the

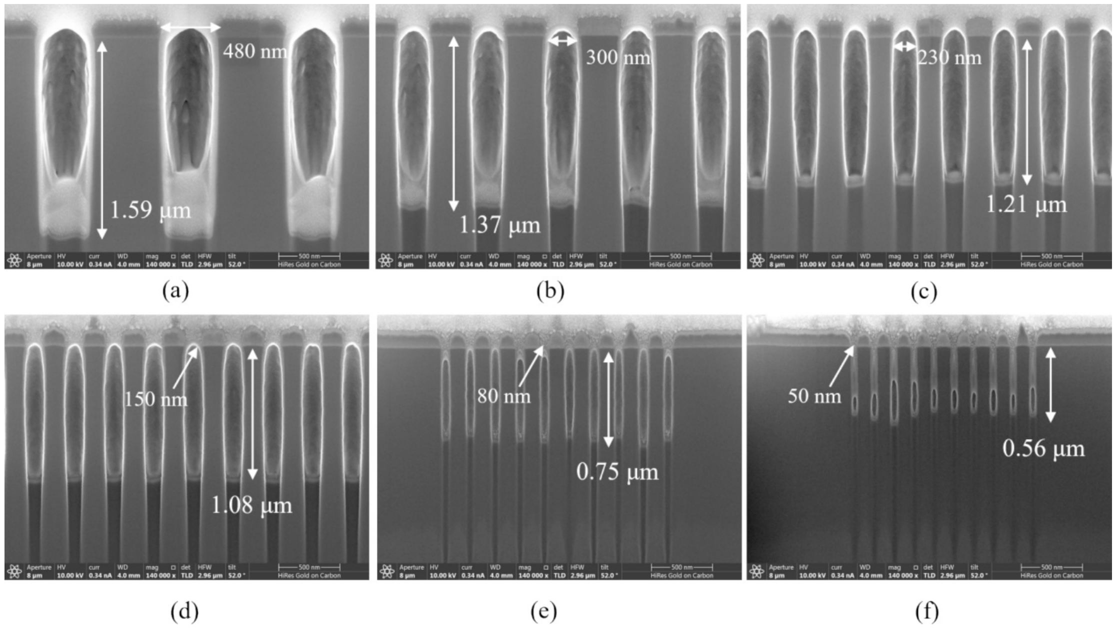

Fig. 13. Etching of different diameter nano-TSVs using optimized Bosch technology.

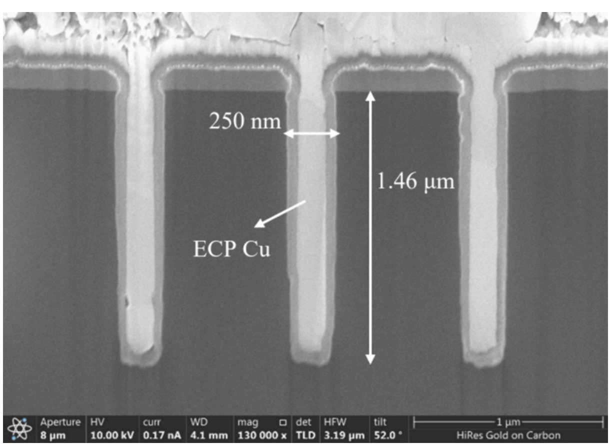

Fig. 14. The results of the metal filling process for nano-TSV using the optimized Bosch technique.

nano-TSV is successfully filled using electroplating filling technology, which is applicable to BSPDN technology and can also be used in 3D IC, 3D NAND, CIS and other technologies.

Declaration of competing interest

The authors declare no conflicts of interest.

Data availability

Data will be made available on request.

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